Advance Preparation

Using a reference such as the IAEA Live Chart of Nuclides or the Radiological Health Handbook, find the radiations and energies expected from natural thorium in equilibrium with its decay daughters.

Introduction

The objective of this experiment is to determine the characteristics of proportional counters. You will use a windowless 2π gas-flow counter with P-10 gas. Avoid initiating a continuous discharge at high voltage, as indicated by an abrupt increase in the observed count rate. **Every time you increase the high voltage you must observe the change in count rate on the counter so that you can reduce the high voltage immediately if excessive count rates are observed.** Note that you should expect a large difference in amplitude between the pulses from alpha particles and beta particles.

Make sure you save all spectra from MAESTRO32, in .SPE format. These files can be read by standard spreadsheet software such as Microsoft Excel and other data analysis software.

Information

- Name:
- Lab Station:
- Lab Partner:
- Date:

1 Pre-Lab Conceptual Questions

1. Draw the expected counting curve for a proportional counter when exposed to both an alpha and a beta source.
2. Draw a plot of the log of pulse amplitude vs. applied voltage for gas counters with applied voltages from 0-2000V for two different energies of incident radiation. Explain the curves, especially the plateaus and the linear regions.
3. With a 2π proportional counter, what is the expected absolute efficiency of the device when counting alpha or beta emitters?
4. Draw the expected differential pulse height spectrum when the detector is exposed to a monoenergetic alpha source.
5. Explain the process of quenching in proportional counter and GM counter. How are they different?

2 The Counting Curve

2.1 Equipment Setup